

D. Y. Patil College of Engineering and Technology, Kolhapur
Department of Computer Science & Engineering

Class: SY-A Subject: AOOC

Experiment no: 15

Group No. 24

Mini Project

Title of Mini-Project: AI Chatbot using Java and Ollama

Problem Statement:

With the rise in popularity of AI-based conversational agents, many existing solutions rely heavily on cloud-based APIs. This project aims to build a lightweight, locally hosted AI chatbot interface using Java Swing for GUI and Ollama (a local LLM server) as the backend. The goal is to allow users to interact with a smart chatbot in real-time without sending data over the internet.

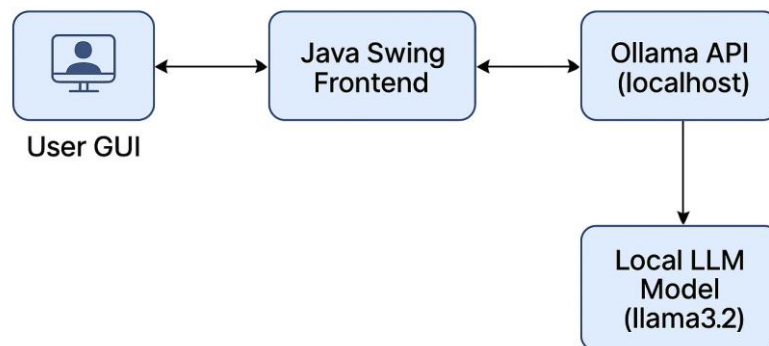
Introduction:

Chatbots powered by large language models (LLMs) are becoming integral in customer support, education, and productivity tools. However, most implementations require a constant internet connection and depend on external services, which may not be suitable for privacy-sensitive or offline environments.

This project demonstrates a simple desktop-based AI chatbot built in Java, utilizing a local Ollama server hosting the llama3.2 model. It features a clean graphical interface with real-time interaction and streaming response handling.

System Architecture:

- **Frontend (Java Swing):** Handles UI components and user input/output display.
- **Backend (Ollama):** Receives user prompts through HTTP POST, processes using a local LLM, and streams output.
- **Communication:** Performed using Java's HttpClient with streamed JSON parsing for real-time interaction.



System Architecture

Module description or working of system:

1. User Interface Module:

- Built using Java Swing.
- Components:
 - JTextField for user input.
 - JButton to send the message.
 - JPanel with scrollable chat area.
 - Chat messages styled differently for user and bot.

2. Message Handling Module:

- Captures user input and sends it to the backend.
- Disables input fields during processing to prevent multiple submissions.
- Adds user message to the GUI immediately.

3. API Communication Module:

- Constructs a JSON request to the Ollama API.
- Sends the message via HTTP using Java's HttpClient.
- Streams the response line-by-line to allow real-time feedback.
- Parses the "content" field from JSON chunks and updates the bot's message accordingly.

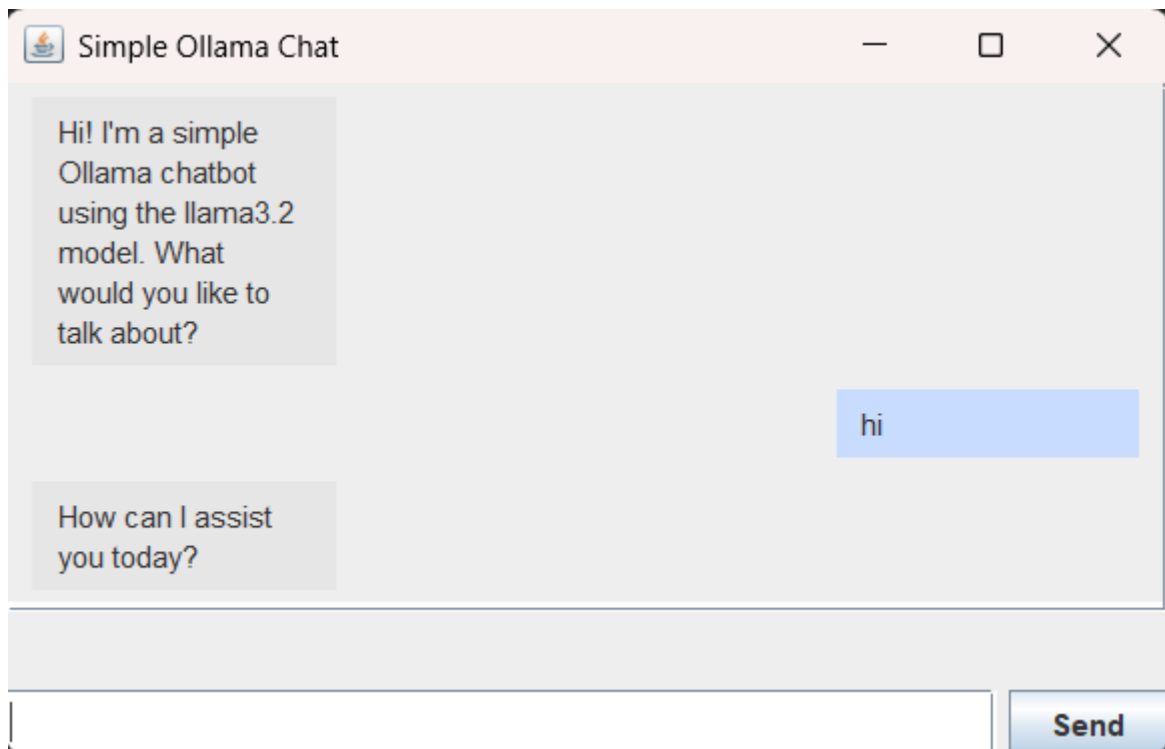
4. Ollama Backend (External):

- Locally runs an LLM server (e.g., llama3.2).
- Listens on localhost:11434/api/chat.
- Receives prompt and returns streamed responses formatted in JSON.

5. Response Rendering Module:

- Processes the streamed output.
- Appends each part of the bot's response to the GUI dynamically.
- Re-enables input controls after response is complete.

Screenshots:



Group Members:

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